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Heritage Impact Assessment, beyond an Assessment Tool: A comparative analysis of urban development impact on visual integrity in four UNESCO World Heritage Properties



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ABSTRACT

Urban growth and modernisation are inevitable phenomena in the developing and developed countries, in which, cultural heritage properties as sensitive urban components may encounter irreparable losses. In the context of UNESCO World Heritage, the uprising conflict between urban development policies and heritage conservation in recent years has drawn more attention to Heritage Impact Assessment (HIA) as an assessment tool to identify and analyse human-induced impacts on cultural heritage properties by the aim of maintaining a balance between cultural heritage protection and urban development needs towards sustainability. This paper aims to review the consequence of urban development pressure on cultural World Heritage properties. It further aims to explore the role of Heritage Impact Assessment (HIA), as an assessment instrument, in recognising and analysing the potential threats in advance and in solving the potential conflicts to safeguard the heritage values. The challenge of conservation of cultural heritage within the broader urban development policies is conducted through reviewing the relevant reports and guidelines. Furthermore, a comparative analysis of four World Heritage properties, that their visual integrity has been affected by urban development projects, is carried out to show the different procedures in mitigating the impact and the role of HIA in the processes. Respectively, this paper compares two World Heritage properties of Meidan Emam (Esfahan, Iran) and Cologne Cathedral (Cologne, Germany) which faced a similar urban development threat at the same time. It further elaborates on how the state parties responded to the identical uprising conflict differently. Moreover, the two cases of Golestan Palace (Tehran, Iran) and Historic Centre of Vienna (Vienna, Austria) are reviewed to emphasise the need for multi-sectoral cooperation and capacity-building in the area of HIA, beyond its assessment methodology, to adequately manage the urban development pressure in the future.

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1. Introduction

While urban features have constantly changed and developed by growing the population, cultural heritage, as an integral part of urban areas, has continued to express the historical evolution of cities. In fact, protection and sustainable development of cultural heritage as a source of identity have a key role in keeping the link between past and future. However, inter alia due to the phenomenon of rapid urbanisation, it should be taken into consideration that heritage properties and their values might

be threatened by new developments in the urban areas; such a gap between cultural heritage conservation and a need for urban development has brought various challenges in the historic urban environment.

Admittedly, the management of World Heritage properties in urban areas needs an appropriate assessment tool for mitigating the potential impacts of new constructions. A high percentage of cultural properties that have been affected by urban development represents this critical threat and source of conflict. Some examples of World Heritage properties affected by new urban development are the conflicts of the Golden Horn Bridge in Istanbul, Turkey from 2006 until 2011; “Ohkta Centre” skyscraper in the vicinity of the historic centre of Saint Petersburg, Russia in 2008; Dresden Elbe Valley in Germany due to the construction of Waldschlösschen Bridge from 2005 until 2009; and Liverpool city centre (UK),

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caused by 30-storey high-rise constructions since 2012. Recognising the challenge, Heritage Impact Assessment is a new assessment instrument to identify and analyse the potential impacts of human-induced threats and projects on cultural World Heritage properties.

2. Research aim

This paper aims to review the consequence of urban uprising projects and urban development threats on cultural heritage properties. Subsequently, the need of Heritage Impact Assessment as a conflict solving tool to enhance the cultural heritage conservation will be discussed. Four World Heritage properties of Meidan Emam (Esfahan, Iran), Cologne Cathedral (Cologne, Germany), Golestan Palace (Tehran, Iran), and Historic Centre of Vienna (Vienna, Austria) are investigated to explore how state parties differently responded to the urban construction threats, and whether HIA could help reduce the impact and speed up the conflict solving process.

3. Theory and background

3.1. Cultural heritage and urban development threats

Articles 2 and 13 of the Convention on the Protection and Promotion of the Diversity of Cultural Expressions [1] emphasised the role of culture in sustainable development. Subsequently, the United Nations conference on sustainable development 'Rio + 20' [2] had a significant role to integrate the culture as a part of the sustainable development that can facilitate safeguarding heritage values within the urbanisation process. Furthermore, the conclusions on cultural heritage as a strategic resource for a sustainable Europe ([3], p. 2) stresses the role of cultural heritage as "a powerful driving force of inclusive local and regional development" as well as the potential resource in sustainable urban development and rehabilitation.

Due to the significant role of cultural heritage in sustainable development, there are various considerations for heritage protection within the urban development-related policies and conventions. For instance, the EU commotion's key action plan Cities of tomorrow and cultural heritage with the aim of "improving damage assessment on cultural heritage, developing innovative conservation strategies, fostering the integration of cultural heritage in the urban setting" [4]. Moreover, United Nation (UN) 2030 Sustainable Development Agenda, Goal 11 (2015, p. 8) calls for "make cities and human settlements inclusive, safe, resilient and sustainable"; the target 11.4 underlines obviously 'Heritage' where declares "strengthen efforts to protect and safeguard the world's cultural and natural heritage" ([5], p.26).

Importantly, based on a static analysis of State of Conservation (SOC) reports prepared by UNESCO ([6], p. 12), "between 1979 and 2013, 66% of the [damaged] properties reported were cultural properties" this properties affected by "Buildings and Development" factor which is not comparable with natural and mixed properties and highlights the need for more attention to the cultural heritage [6].

Although "cities are increasingly seen as drivers of sustainable development by applying sustainability targets" [7], the urban development strategies and projects have remained one of the main threats to cultural heritage values. The State of Conservation (SOC) reports from 1979 till 2013 [6] consider the fact that "Building and Development" has the most common threat to CH properties after "Management and institutional Factors". It is also worth noting that "Transportation Infrastructure" is located at the top of the list in this period (Ibid).

Even though each case study has its characteristics and the impacts could be different from case to case, some common negative impacts due to urban development projects can be recognised.

One of the main adverse effects of urban development, which is mentioned in UNESCO's State of Conservation Report (SOC), is the impact on the visual integrity and aesthetic values of cultural properties as a result of high-rise buildings [8].

Recently, the visual impact also is addressed as an important factor that should be assessed in EIA in order to protect cultural heritage. For example, Directives 2014/52/EU of the European Parliament and the Council on the assessment of the effects of certain public and private projects on the environment (2014) indicates that "to better preserve historical and cultural heritage and the landscape, it is important to address the visual impact of projects, namely the change in the appearance or view of the built or natural landscape and urban areas, in environmental impact assessments."

The increasing number of cultural properties that have been affected visually by high-rise buildings as well as the new incompatible constructions in the urban areas (e.g. Dresden, Liverpool, London, Seville, and Vienna) emphasises the challenge.

3.2. Development of Heritage Impact Assessment (HIA) in the Framework of Environmental Impact Assessment (EIA)

Since 1970, Environmental Impact Assessment (EIA) has been used to evaluate the impacts of development on the environment as the official tool "to identify the environmental, social, and economic impacts of a project prior to decision making. It aims to predict environmental impacts at an early stage in project planning and design, find ways and means to reduce adverse impacts, shape projects to suit the local environment, and present the predictions and options to decision makers" [9].

Importantly, in order to protect the environment from the adverse effects of a project proposal, EIA should be incorporated into the management system of urban and regional levels. In this regard, IAIA and UK IEA [10] recognised the primary objective for implementation of EIA "addressed and incorporated into the development decision-making process" and "promote development that is sustainable and optimises resource use and management opportunities". Substantively, EIA has emerged as a productive assessment tool to provide a systematic and integrated methodology for examining the project proposals, assessing impacts, and facilitating decision-making by determining mitigation strategies. Moreover, EIA is established as a legally binding instrument in several countries around the world to prevent the adverse effects of development proposals prior to major decisions being taken [11].

In 1999, the International Association for Impact Assessment (IAIA) published the principles for EIA as an interdisciplinary process. In this document, IAIA distinguishes the culture, tradition, and value of communities as features which should be respected and noticed in the Environmental Impact Assessment. Subsequently, in many EIA guidelines, cultural heritage is addressed as a separate and sensitive element that needs to be considered in such a way that its qualitative characteristics can be protected.

As mentioned by UNESCO, Heritage properties are "non-renewable resources" of each community which extremely linked to the identity and sense of belonging. The World Heritage Convention 1972 recognised Outstanding Universal Values (OUV) as a central concept of World Cultural and Natural Heritage which deserves to be protected and transmitted to the next generation. Safeguarding heritage values is a fundamental factor in the management and protection of World Cultural Heritage; and therefore, identifying and mitigating potential threats to the values need to be an integral part of heritage conservation. "Value has always been the reason underlying heritage conservation. It is self-evident that no society makes an effort to conserve what it does not value" ([12], p. 3).

In this regard, state parties are responsible for safeguarding the OUV and its associated attributes, while considering the integrity



Fig. 1. (left) Jahan Nama High-rise Tower (red circle) had impact on visual integrity for Naghsh-e Jahan Square © CHN.ir (2004), (right) two locations of Naghsh-e Jahan Square (green) and Jahan Nama Complex (red) on the google map © 2017 Digitalglobal, Map data © 2018 Google earth and Image 2019 Maxar Technologies/B. Ashrafi.

and authenticity of World Heritage properties from significant negative impacts imposed by changes and developments. Accordingly, EIAs do not explicitly address the OUV “where cultural heritage sections of EIAs do not focus clearly on the attributes of OUV, they would not meet desired standards in managing change at WH properties” [13].

Recognising the challenge, within the context of ‘Environmental Impact Assessment’ (EIA), Heritage Impact Assessment (HIA) has been developed in recent years to consider the specific characteristics and qualitative values of World Heritage properties as a fundamental factor in management and protection of cultural heritage. Heritage Impact Assessment is a procedure to identify and analyse the potential impacts of human-induced threats on cultural heritage, and therefore, it supports better protection and management of heritage assets. The need for development and change on one side, and heritage conservation, on the other side, may cause conflict among stakeholders dealing with new development in historic urban areas. Heritage Impact Assessment is a tool to fill the gap between the demands of development and heritage conservation. Furthermore, HIA provides a new assessment tool to facilitate the cultural heritage conservation within the broader sustainable development policies.

4. Methods: comparative analysis of four case studies

Following the clear understanding of the role of HIA in identifying and analysing the potential threats in advance and in reducing the conflicts to safeguard the heritage values, four concrete case studies are reviewed. The case studies are tangible examples to compare the urban development threats in the different urban environments in three phases of pre-, during, and post-construction. Importantly, two WH properties of Meidan Emam and Cologne Cathedral were chosen due to the similar urban development threat at the same time to demonstrate how the state parties responded to the identical uprising conflict differently. Moreover, the recent examples of Golestan Palace and Historic Centre of Vienna are studied to emphasise the necessity for multi-sectoral cooperation and capacity-building in the area of HIA with the purpose of better managing of the same problem in the future.

This study considered both primary and secondary information. The HIA reports and nomination dossiers as well as UNESCO World Heritage Committee decisions and ICOMOS Advisory Body evaluations were reviewed to support the quality of the analysis. Besides, the primary data, which was gathered through onsite observation and photography of the heritage properties, was also cross-checked with the data extracted from the official records.

4.1. Meidan Emam (Naghsh-e Jahan), Esfahan, Iran

Esfahan (Isfahan) as a major city in Iran has been developed and expanded in different eras. The city was the capital of Iran, two times in its history, and encompasses a vast number of valuable historic buildings, gardens, and streets which represent the aesthetic value of Iranian-Islamic architecture and urban planning. Since the 19th century and in particular recent decades, the historic city of Esfahan has been facing the modernisation process including numerous new urban development projects to respond to the current needs of the community. Therefore, the city has faced a challenge to find the proper solutions and keep the balance between the protection of historic-cultural properties and inhabitants' demands of urban development.

One of the recent conflicts in this regard is a construction of high-rise building (Jahan Nama Tower) in the vicinity of the World Heritage property of Naqsh-e Jahan Square (Fig. 1 right) in Esfahan (700 m distance). Naqsh-E-Jahan (Emam) square is the most well-known cultural property in Esfahan as well as Iran which was built in the 17th century and inscribed as a World Heritage List in 1979 under criteria (i)(v)(vi). Naqsh-e Jahan is one of the largest squares in the world surrounded by unique Iranian-Islamic architecture and considered to be “an impressive testimony to the level of social and cultural life in Persia during the Safavid era” [14].

In 1997, Esfahan municipality permitted the construction of this 48-metre multifunctional complex to a private sector. This new structure height was beyond the building code limitation and was entirely visible from the square (Fig. 1 left). Consequently, it had an enormous adverse impact on the skyline of the Esfahan as well as the visual integrity of the Naghsh-e Jahan. In 2002, the World Heritage Committee requested the Iranian government to minimise the negative impact of the Jahan Nama commercial complex for the first time ([15] WHC-02/CONF.202/25).

In 2003, the Iranian government agreed to reduce the height of the building as well as provide the necessary funds. Two years later, in 2005, the destruction commenced based on the height level determined by UNESCO, which stated that 12-metre height in the eastern side and 24-metre in the western side should be reduced in order ([16] WHC-05/29.COM/22) “to minimise the negative impact this complex causes upon the setting, function, and integrity of the Meidan Emam World Heritage property and Historic City of Esfahan” ([17,18], p.112, WHC-04/28.COM/26). However, until 2006 only one store of the building was removed. Due to the slow process of reducing the Jahan Nama height, the World Heritage Committee sounded the alarm that the property will be inscribed on the List of World Heritage in Danger.

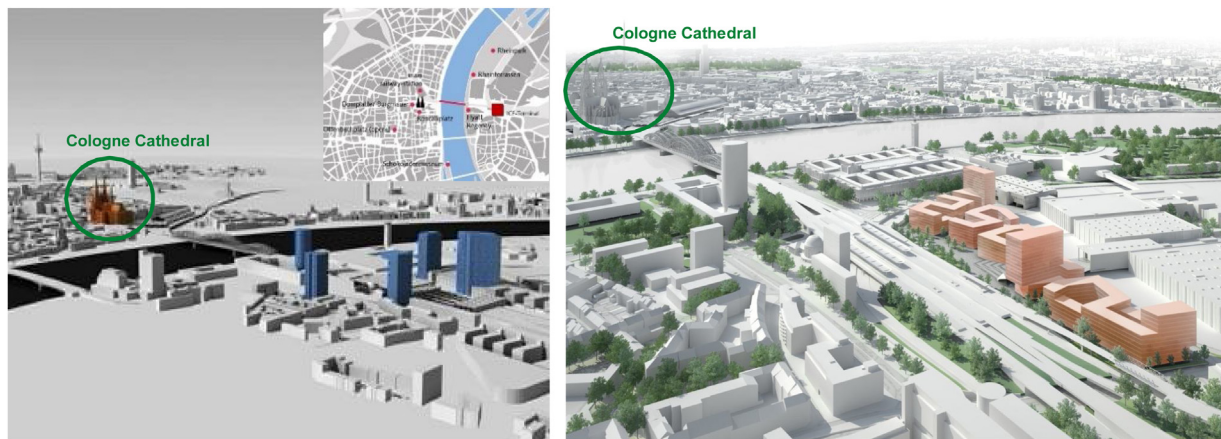


Fig. 2. (left) Original planned ICE High-rise Cluster in blue (right) the new adapted scheme in respect to OUV of the Cologne Cathedral in red © v-cube/HH Vision/STRABAG-ECE cited in Institute of Urban and Regional Planning, RWTH Aachen University.

Besides, in 2008 the plan for a subway construction (line 2) which crosses through the Esfahan's historical axis faced the Meidan Emam a new urban development threat, and following in 2009, UNESCO asked the authorities to conduct the Environmental Impact Assessment (EIA) for the planned subway line ([19] WHC-09/33.COM/20). Moreover, the demolishing progress of the Jahan Nama commercial complex faced financial and structural problems. It took until 2014, to complete the height reduction of the tower. Finally, after more than 11 years, the Jahan Nama commercial complex had been adjusted to a proper height level.

This case was a difficult experience for the Iranian state party and the city of Esfahan. Nevertheless, the World Heritage Committee stated that “it may appear that the Jahan Nama building demolition has been a painful process but the State Party authorities have learned lessons from this case. The completion of the demolition work also shows the increased awareness of the authorities for the preservation of the cultural values of the World Heritage property in a wider setting.” ([20], WHC-13/37 COM/7B.Add.)

4.2. Cologne Cathedral, Cologne, Germany

In parallel to the case of Meidan Emam and the Jahan Nama commercial complex, Cologne as one of the major cities in North-West Germany with a population of approx. one million inhabitants developed a new plan of high-rise tower blocks in 2002. Shortly after, the UNESCO World Heritage Committee raised concerns about the possible negative impacts of this plan on the historical core zone of the city and especially Cologne Cathedral, inscribed in the UNESCO World Heritage List in 1996. The project included five high-rise tower blocks with more than 100-m height which was supposed to be located on the east of the Rhine river (Deutz district). Due to the short distance of the project site, UNESCO was concerned about adverse visual impacts would occur on both the historic skyline of the city and particularly on Cologne Cathedral.

Cologne Cathedral, one of the best-known sites in Germany and worldwide, was inscribed on the UNESCO World Heritage List under criteria (i)(ii)(iv). This Gothic masterpiece “bears witness to the strength and endurance of European Christianity. No other Cathedral is so perfectly conceived, so uniformly and uncompromisingly executed in all its parts”. [21]

In 2003, UNESCO World Heritage Committee ([22] WHC-03/27.COM/24) demanded a detailed report of the newly planned urban development project from the German authorities. Following in 2004 (WHC-04/28.COM/26), the property was inscribed on the List of World Heritage in Danger due to a lack of enough information and collaboration of the state party.

Consequently, in 2005 the construction project in the vicinity of the Cathedral was stopped by Cologne municipality and a visual impact study has been compiled by RWTH Aachen University. The study indicated the high negative impacts of the proposed high-rise tower blocks on the Cologne Cathedral and its setting [23]. Correspondingly, the assessment recommended, “the ICE High-Rise Cluster should be reviewed and the buffer zone around Cologne Cathedral should be enlarged by a monitored zone to avoid similar conflict in the future”. (Ringbeck, 2006, Kloos et al., 2005 cited by [24]).

Finally, after several workshops and conferences at different levels, the planned high-rise project was adapted to a new more compact urban development scheme with a lower height concerning the visual integrity of Cologne Cathedral (Fig. 2). Besides, the buffer zone extension was confirmed by the City of Cologne to protect the World Heritage site from further adverse urban development. Subsequently, Cologne Cathedral was removed from the List of World Heritage in Danger in 2006 ([25] WHC-06/30.COM/19), and further on, the buffer zone was modified in 2008. Furthermore, [26] in their evaluation in 2008 mentioned that the new buffer zone “meets concerns relating to the visual integrity and urban cultural landscape of the cathedral. It covers an urban surface area of 258 hectares.”

4.3. Golestan Palace, Tehran, Iran

Tehran has been the capital city of Iran since 1796 (Qajar dynasty), and it contains numerous historic sites and cultural properties conveying the heritage of the city and its associated community. Golestan Palace is a unique property which has shined in all life of Tehran and was inscribed on UNESCO's World Heritage List in 2013 under criteria (ii)(iii)(iv) [27]. The palace is located in the old Arg (citadel) and central zone of the town, as well as in the vicinity of the Grand Bazaar of Tehran, one of the most crowded parts of the city.

Tehran as a major and the capital city has faced manifold new urban developments and renewal projects in recent years. The old downtown and Bazaar area have faced a great interest regarding urban development, where the old buildings have been replaced by modern structures with the great financial interest, and a huge number of people commute every day. Therefore, this strategic district has gasped public and private investors, but the high rate of new development and construction could considerably threaten Tehran's historic centre and the cultural properties in the area.

Although ICOMOS (2013) mentioned in its evaluation report that “the delimitation of the buffer zone does not cover a large



Fig. 3. (left) Location of a new shopping mall (red) on the boarder of buffer zone of the Palace based on nomination dossier in 2013 © google earth (2019) and Image 2019 Maxar Technologies/B. Ashrafi, (right) a view of the new construction from the palace entrance © chekhabaronline.ir, photo by A. Mirzazadeh.

enough area to provide for the long-term protection of the property's setting", no adequate actions for expanding the buffer zone have been undertaken. Besides, ICOMOS stated the negative impact of a recently built 6-storey building on the east side of the Palace which "the State Party stated that processes were underway to reduce the height of this building." [28]. Up until now, the area still encloses many new urban development projects.

One of these projects is Saraye Delgosha, which was the historic caravansary built in the Qajar dynasty (more than 130 years old) and listed as a national heritage in Iran. Unfortunately, this historic building was sold in 2010 after delisting from the Iranian heritage list, and the investors were permitted by the Tehran municipality to build a shopping mall with a total height of 27-metres [29].

Subsequently, the Golestan Palace was nominated for inscription in the World Heritage List, and the concerns grew about this high-rise building. According to the nomination dossier of the palace (2012), the maximum allowable height of new constructions in the buffer zone is 7.5 metres, while the height in the Bazaar area is limited to 10.30 metres regarding Tehran Municipality regulations. However, the newly planned shopping mall is located only about 9 metres distance from the buffer zone and approximately 122-metre from the entrance gate of the palace. Hence, it was obvious at that time that this building would have a negative visual impact on the World Heritage property. In 2013, after long discussions between ICHHTO (Iranian Cultural Heritage, Handicrafts & Tourism Organisation), municipality, and private owners, it was decided to construct a 5-story building as well as retreating this building 30 metres from the main street. Nevertheless, it will still exceed the building code in this district. At least 1.5 stories of the building should be demolished to reach a proper building code (Fig. 3).

In conclusion, it has to be stated that Iran faced a serious challenge to cope with the negative impact of high-rise buildings on the World Heritage property, Naghsh-e Jahan in Esfahan; currently, the Golestan Palace as the only World Heritage property of the capital city faced the same threat. Up until now, the state party has not yet conducted a HIA to assess the potential negative impacts of these projects on the World Heritage properties. Consequently, no concrete solutions and mitigation measures have been achieved.

4.4. Historic Centre of Vienna, Vienna, Austria

The analysis of the comparable cases in Vienna shows similar effects. "Historic Centre of Vienna" was inscribed as a UNESCO World Heritage property under criteria (ii), (iv), and (vi) in 2001

(Decision 25 COM XA). As mentioned in nomination dossier, the historic city developed in three major phases of "Mediaeval, Baroque and the Gründerzeit" (Criterion iv), which characterise both Vienna's skyline and many famous vistas to Vienna's historic city centre. Despite those several development periods, these vistas were always respected so that Vienna's historic silhouette was considered to be largely intact when the property was inscribed in the World Heritage List. However, Vienna's more recent urban development shows that the city has continuously been changing and rebuilding due to the demand of its rising population. Consequently, Vienna has been facing several challenges concerning its new urban development since its inscription, especially in the vicinity of the historic city centre. The site has been threatened by several high-rise projects with a negative visual impact such as Wien Mitte/Zollamtsviertel (2002), Kometgründe (2007), Vienna Central Train Station (2008), and Heumarkt Neu (2011). In 2008 and 2009, ICOMOS and UNESCO World Heritage Committee requested "a comprehensive visual impact assessment of the entire project" due to the high-rise constructions of Wien-Mitte and Vienna Central Train Station. Accordingly, a Visual Impact Study was compiled, and Austrian authorities reduced the 100m height of the tower to 88m. Following, the restricted maximum height of 60m was recommended by the WH Committee and ICOMOS in 2012.

These questions culminated in recently planned high-rise project Heumarkt Neu. This project has been subjected to controversial discussions since 2011. The concept of the project is the result of an international architectural competition organised in 2014 and includes the extension and redevelopment of the Inter-Continental Hotel and foresaw an upgrade of the public spaces surrounding the Konzerthaus, and the so-called Ice-skating Club. As a result, the winning design proposal exceeded the height restriction recommended by ICOMOS International in 2012 (Fig. 4).

Both ICOMOS and UNESCO World Heritage Committee have addressed frequently the negative impacts of the Heumarkt Neu project on the visual integrity of the World Heritage property. Moreover, the famous view from the unique Baroque garden of the Belvedere on the historic skyline and townscape of Vienna was especially a matter of concern (Fig. 5). Due to the lack of modifications in the proposed design proposal, the property was inscribed on the List of World Heritage in Danger in 2017 (decision 41 COM 7B.42).

In 2019, the Federal Chancellery of Austria commissioned a "Heritage Impact Assessment on the Heumarkt Neu Construction Project and Development of the World Heritage property Historic



Fig. 4. (left) The main property and buffer zone of Historic centre of Vienna and location of Heumarkt Neu project © Republic of Austria. (2000)/B. Ashrafi. (right) Comparison and overlapping of the existing building (red) and planned Heumarkt Neu project (blue) © building models: Isay Weinfeld & Sebastian Murr/city model: MA41/Installation: v-cube.

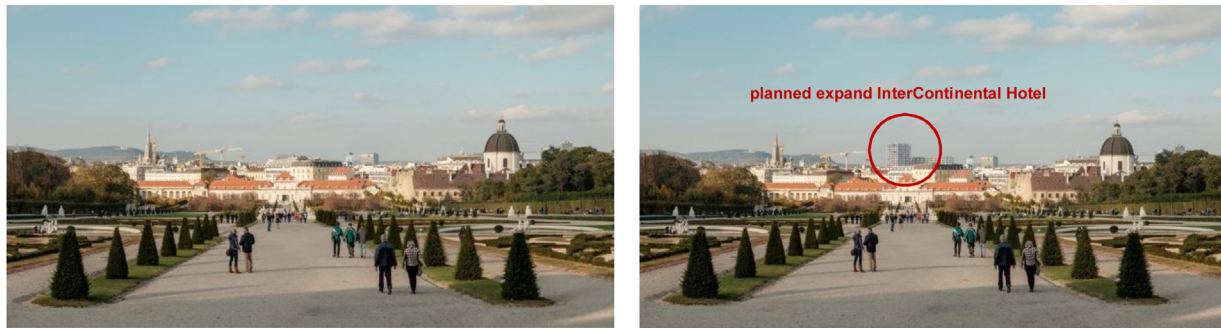


Fig. 5. (left) current situation of Belvedere Garden view including Stephan's Cathedral © michael kloos planning and heritage consultancy, 2019, (right) planned expand InterContinental Hotel © michael kloos planning and heritage consultancy/v-cub, 2019.

Centre of Vienna". To provide a basis for this study, the attributes of the World Heritage property and characteristics of Vienna's skyline were defined to identify the relevant viewpoints which convey the OUV of the site. Afterward, the Heumarkt Neu project was visualised from these viewpoints to assess its impact on the OUV of the World Heritage property.

Based on the assessment, the proposed Heumarkt Neu project was considered to have large/very large impacts on the various attributes of the Historic Centre of Vienna (Chart 1). Afterward both the authors of the study and ICOMOS International recommended stopping the project for two years to develop better protection measures concerning the OUV. In particular, it was recommended to compile a Management Plan, thus "adding formal protection and planning instruments at the level of the city of Vienna" [30].

Meanwhile, the Municipality of Vienna decided to start the works on this Management Plan and structural changes on the Heumarkt Neu project are discussed. Recently, plans to transform the Heumarkt Neu project by replacing the planned tower block were discussed and a stronger legislative framework to protect Vienna's World Heritage properties is at least considered by the Municipality.

5. Discussion and results

According to the World Heritage Convention 1972, each State Party is responsible for protecting its cultural and natural World Heritage properties. In the case of Meidan Emam, Esfahan in Iran, the state party did not recognise the threat or did not act in time, and therefore, UNESCO was informed too late when the construction of the tower had been mostly carried out. Although the Iranian government started its collaboration with UNESCO regarding minimising the adverse impacts of the high-rise building on the visual integrity of the WH property, there was no adequate methodology

of impact assessment conducted to adequately identify and analyse the existing threats and associated impacts on the site. According to the ICHHTO interview in 2010 (cited by [31]), the long process of reducing the building height costed more than 3 million dollars for the government. This money was needed not only for the demolition of extra levels of the tower but also as compensation for the owners.

Concerning the case of Cathedral Cologne, the state party did also not cooperate appropriately with UNESCO at the first step; and this led to placing the property on the List of World Heritage in Danger. However, the compilation of the impact assessment resulted in both an improvement of this cooperation of all stakeholders as well as in a suitable solution considering the visual integrity of Cologne Cathedral. Besides, the assessment resulted in the extension of the buffer zone, which provides the Cathedral and its setting with more strict protection regulations.

The timeline chart (Fig. 6) clearly maps the entire process of emerging threats and UNESCO – State Parties communication to reduce the negative impacts on the two cases of Esfahan and Cologne. Even though both cases faced a similar problem in the same period, they experienced different procedures to tackle the conflict. In the case of Cologne Cathedral, HIA provided suitable mitigation strategies as well as speeding up the mitigation process through communication and collaboration between different stakeholders. In the case of Esfahan, however, the lack of conducting a HIA resulted in a far longer process and considerable costs. Although the problem has been solved after 16 years at the international level, there are still many unsolved conflicts between the local authorities and the private sector. Besides, there are still no recommendations and regulations for further urban development, and up to now neither any suggested buffer zone for the WH property to avoid nor at least identify the further threats in advance.

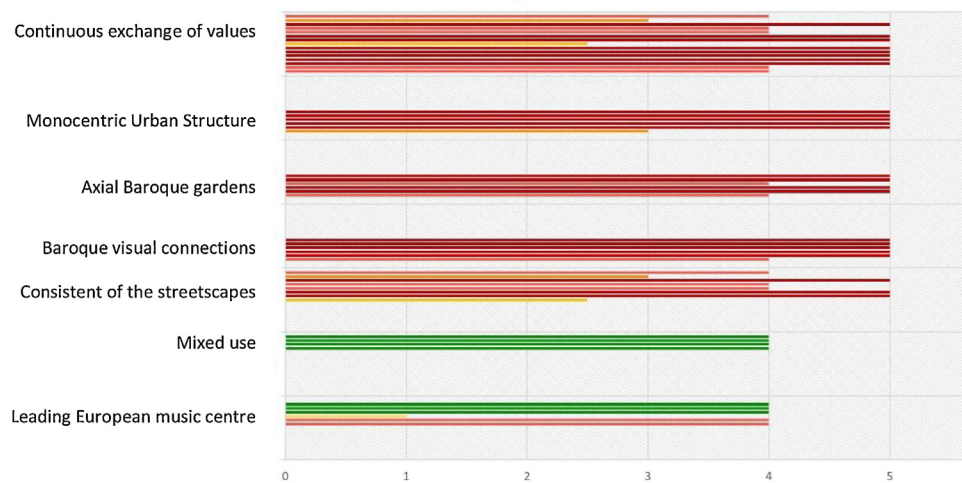


Chart 1. Effects on attributes of the World Heritage property Historic Centre of Vienna (© Baharak Seyedasrafi, 2019 cited in [30]). According to the ICOMOS HIA guideline 2011, the magnitude of impacts ranks in five degrees, from 'major' to 'no change'. This chart was produced based on the ICOMOS guide for assessing severity of Impacts which is developed and published by [11]. In this study magnitude of the negative impacts is measured by 5 scale-point from Sever to Negligible with 5 reddish colour codes for negative impacts and 5 greenish colour codes for positive impacts. The vertical line in the chart represents the different attributes of the Historic Centre of Vienna and the horizontal line shows the magnitude of impacts.

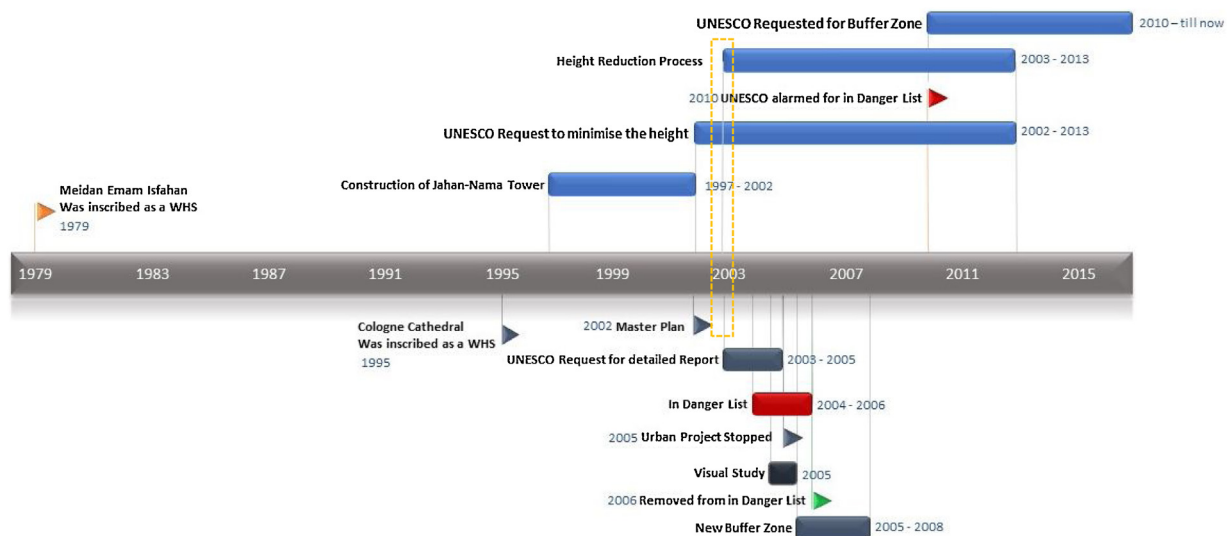


Fig. 6. The – State Parties communication for two affected World Heritage Properties of Meidan Emam Esfahan and Cologne Cathedral by urban development projects (B. Ashrafi).

While UNESCO stated that the challenges faced in the case of Esfahan were “a good lesson for the Iranian government” and has raised their awareness, the repetition of the recent large-scale construction in the vicinity of Golestan Palace as the only World Heritage property of the capital city of Tehran reveals clearly that similar threats continue to affect Iranian World Heritage properties.

Similar to Iranian cases, the study of Vienna's urban development demonstrates the current challenges between urban development and cultural heritage protection. Although the case of Vienna is not yet solved completely, Vienna municipality is working currently on a new Management Plan as recommended in the HIA. The case of Vienna represents the fact that HIA without adequate mitigation strategies could not be an effective assessment tool. These two cases correspondingly determine that having experience is not enough to cope with further challenges and threats, but HIA should be communicated through the communication and capacity-building within multiple sectors to engage in cultural heritage conservation and urban planning.

Through the analysis of the case studies in three phases of pre-, during, and post-construction, it reveals that besides the importance of HIA methodology, there are some factors, such as legislation and multi-sectoral collaboration, mitigation, and implementation actions, and communication and capacity-building in HIA, that significantly contribute to the effectiveness of HIA (Fig. 7). As the figure illustrates there are interrelations between the essential factors. Such an adequate methodology of impact assessment in line with legislation and multisectoral collaboration would improve possibilities to provide mitigation strategies and following-up the implementation of suggested measures. Importantly, the development of the communication and capacity building would raise the effectiveness of HIAs through strengthening the stakeholders' perception, knowledge, and skills with the aim to sustain the World Heritage properties which exposed to future development threats.

It should be noted that a buffer zone is a current practical tool for the protection of World Heritage properties and their setting. However, the cases of Cologne and Vienna clearly show that buffer zones

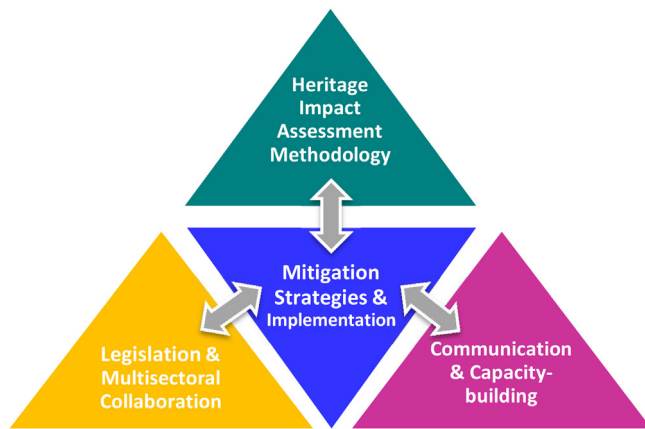


Fig. 7. Various necessary components to enhance effectiveness of HIAs (B. Ashrafi).

are insufficient, especially if only the horizontal dimension is taken into account and visual relationships exceeding the buffer zones are not considered. Besides, there are barely State Parties which anchored buffer zones as a valid protection instrument in their legislation, a fact which weakens their role as protection instrument considerably. In this context, Heritage Impact Assessments can serve as an instrument to supplement buffer zones and they can also support a proper definition of their borderlines. The case of Golestan Palace shows that this World Heritage property still suffers from an inappropriate buffer zone due to lack of considering the visual integrity in its definition of the protecting area, thus leading to a negative impact of the high-rise construction outside of the buffer zone on its visual integrity. However, in the case of Cologne Cathedral the visual impact assessment supported the better protection of the World Heritage property through the extension of the existing buffer zone as an adequate mitigation strategy and to prevent further comparable problems.

Comparing the four cases stresses that HIA is far more effective as a conflict solving instrument if it is applied as early as possible through pro-active approaches. It can help to avoid serious conflicts and costly mitigation measures such as in Esfahan. As a result, there is an essential need for improving the legislative and legal framework for HIA, which subsequently should be embedded in legislation of state parties to the Convention.

6. Conclusion

As mentioned earlier, uncontrolled urbanisation has appeared to be one of the major threats to World Heritage properties. In such a situation, a conflict emerges between urban development and heritage protection policies, particularly in historic urban areas. Heritage Impact Assessment (HIA) has been recently established to specifically manage the challenges related to the loss of heritage values. In order to explore different experiences of cultural properties affected by urban projects, four World Heritage properties of Meidan Emam (Esfahan, Iran), Cologne Cathedral (Cologne, Germany), Golestan Palace (Tehran, Iran), and Historic Centre of Vienna (Vienna, Austria) have been reviewed. The impact on visual integrity has been particularly addressed in this comparative analysis.

The comparative analysis of these four case studies explored the role of HIA as an assessment tool to share a common understanding of the conflict situation among the relative multiple sectors. It further shows how HIA, beyond the assessment tool, can facilitate determining and efficient implementation of mitigation strategies in the decision-making procedure. The functionality and effectiveness of the tool rely on the appropriate methodology, a

clear definition of their legislative framework, and multi-sectoral collaboration, as well as communication and capacity-building. Accordingly, this paper suggests that HIA should be incorporated into the heritage management plan, as well as, into existing assessment tools such as EIA. Within the urban and regional planning context, this can better ensure a safeguarding of World Heritage properties and a more socio-cultural and economic sustainable urban development.

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